

Electromagnetism 2

(spring semester 2026)

Lecture 4

Electric dipoles

- ❖ Electric dipole moment
- ❖ Electrostatic potential and electric field of a dipole
- ❖ Electric dipoles in an external electric field

Previous lecture

- ❖ Ideal conductors in electrostatic field:
 - ✓ the volume of the conductor is equipotential;
 - ✓ there are no volume charges ($\rho=0$);
 - ✓ in general, there are non-zero surface charges ($\sigma \neq 0$);
 - ✓ just above the surface, $E_n = \sigma/\epsilon_0$ and $E_t = 0$.
- ❖ Laplace equation for the electrostatic potential in the absence of free charges:

$$\nabla^2 \varphi = 0$$

- ❖ The uniqueness theorem: the Laplace equation has a unique solution for any complete set of boundary conditions.
- ❖ The method of image charges: a tool for electrostatic field computations in the presence of conductors.

Electric dipole moment

A charge Q distributed over a volume V , near the origin of a reference frame:

$$Q = \int_V \rho(\vec{r}') dV'$$

By the principle of superposition, the electrostatic potential of this charge distribution:

$$\varphi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} dV'$$

When observed from a large distance ($r \gg r'$),

$$\frac{1}{|\vec{r} - \vec{r}'|} = (r^2 + \underbrace{r'^2}_{\text{negligible}} - 2\vec{r}\vec{r}')^{-1/2} \approx \frac{1}{r} (1 - 2\vec{r}\vec{r}'/r^2)^{-1/2} \approx \frac{1}{r} + \frac{\vec{r}\vec{r}'}{r^3}$$

Coulomb term
(dominates at $r \gg r'$ for $Q \neq 0$)

Therefore

$$\varphi(\vec{r}) \approx \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{r}') dV' \left(\frac{1}{r} + \frac{\vec{r}\vec{r}'}{r^3} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{r} + \frac{\vec{p}\vec{r}}{r^3} \right)$$

Dipole term (the leading term for $Q=0$)

$$\vec{p} = \int_V \rho(\vec{r}') \vec{r}' dV'$$

is the *electric dipole moment* of the charge distribution [unit: C·m].

Electric dipoles

For a system of point charges, $\vec{p} = \sum_i q_i \vec{r}_i$

Electric dipole: a (neutral) system with $Q=0$ and $\vec{p} \neq 0$.

Examples: H_2O , $\text{C}_2\text{H}_5\text{OH}$, HCl molecules ($\vec{p} \sim 10^{-29} - 10^{-30} \text{ C}\cdot\text{m}$).

For $Q=0$, dipole moment *does not depend on the origin of coordinates*.

Proof for a system of point charges:

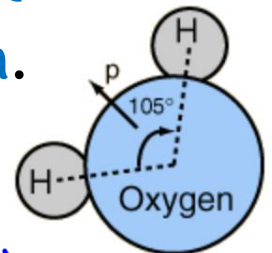
if the origin shifted by $\vec{a}$, coordinates become $\vec{r}_i' = \vec{r}_i - \vec{a}$

Dipole moment for the new choice of origin:

$$\vec{p}' = \sum_i q_i (\vec{r}_i - \vec{a}) = \sum_i q_i \vec{r}_i - \vec{a} \underbrace{\sum_i q_i}_{\text{Net charge, } Q=0} = \sum_i q_i \vec{r}_i = \vec{p}$$

A simple (+q, -q) electric dipole:

two opposite point charges $\pm q$, separated by a distance a .



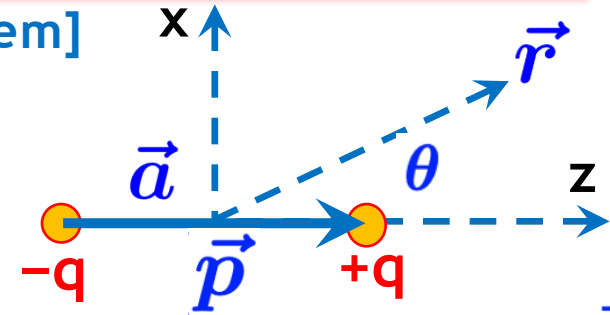
Its dipole moment, using the above definition: $\vec{p} = q\vec{a}$

Potential for a (+q,-q) dipole

[not discussed in the lecture; a non-assessed problem]

Explicit computation for the simple dipole:

using the principle of superposition,
the electrostatic potential is



$$\varphi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{(z - a/2)^2 + x^2 + y^2}} - \frac{q}{\sqrt{(z + a/2)^2 + x^2 + y^2}} \right]$$

At large distances ($|z| \gg a$),

$$(z \pm a/2)^2 + x^2 + y^2 \approx z^2 \pm za + x^2 + y^2 = r^2 \left(1 \pm \frac{za}{r^2} \right)$$

Leading term in the Taylor expansion at large distances ($r \gg a$):

$$\begin{aligned} \varphi(\vec{r}) &\approx \frac{q}{4\pi\epsilon_0 r} \left[\left(1 - za/r^2 \right)^{-1/2} - \left(1 + za/r^2 \right)^{-1/2} \right] \approx \\ &\approx \frac{q}{4\pi\epsilon_0 r} \frac{za}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} = \boxed{\frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3}} \end{aligned}$$

The dipole term, as expected

Dipole electric field

$$\vec{E} = -\nabla\varphi = -\frac{1}{4\pi\epsilon_0}\nabla\left(\frac{\vec{p}\cdot\vec{r}}{r^3}\right)$$

The **z**-component:

$$E_z = -\frac{p}{4\pi\epsilon_0} \cdot \frac{\partial}{\partial z} \left(\frac{z}{r^3} \right) = \frac{p}{4\pi\epsilon_0} \cdot \frac{3z^2r - r^3}{r^6} = \frac{p}{4\pi\epsilon_0} \cdot \frac{3\cos^2\theta - 1}{r^3}$$

Each of the (**x,y**) transverse components:

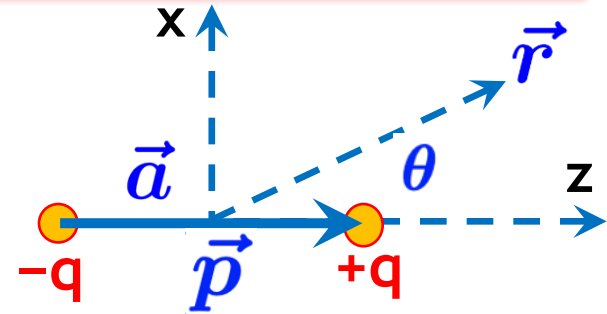
$$E_x = -\frac{p}{4\pi\epsilon_0} \cdot \frac{\partial}{\partial x} \left(\frac{z}{r^3} \right) = \frac{p}{4\pi\epsilon_0} \cdot \frac{3zx}{r^5}$$

The total transverse component:

$$E_{\perp} = \sqrt{E_x^2 + E_y^2} = \frac{p}{4\pi\epsilon_0} \cdot \frac{3z}{r^5} \sqrt{x^2 + y^2} = \frac{p}{4\pi\epsilon_0} \cdot \frac{3\sin\theta\cos\theta}{r^3}$$

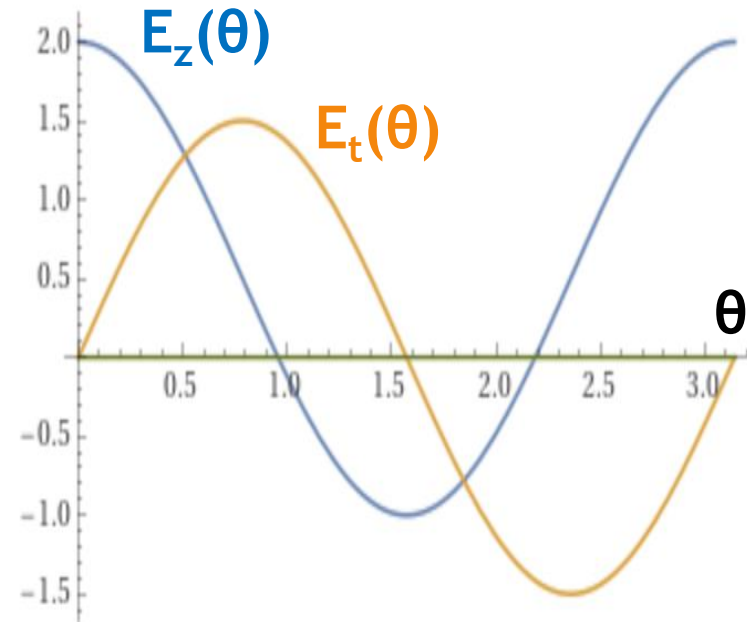
At a fixed direction, dipole field falls off with distance as $\sim 1/r^3$.

Dipole field and potential are symmetric wrt the dipole axis.



Dipole electric field: illustration

The two components of the dipole field

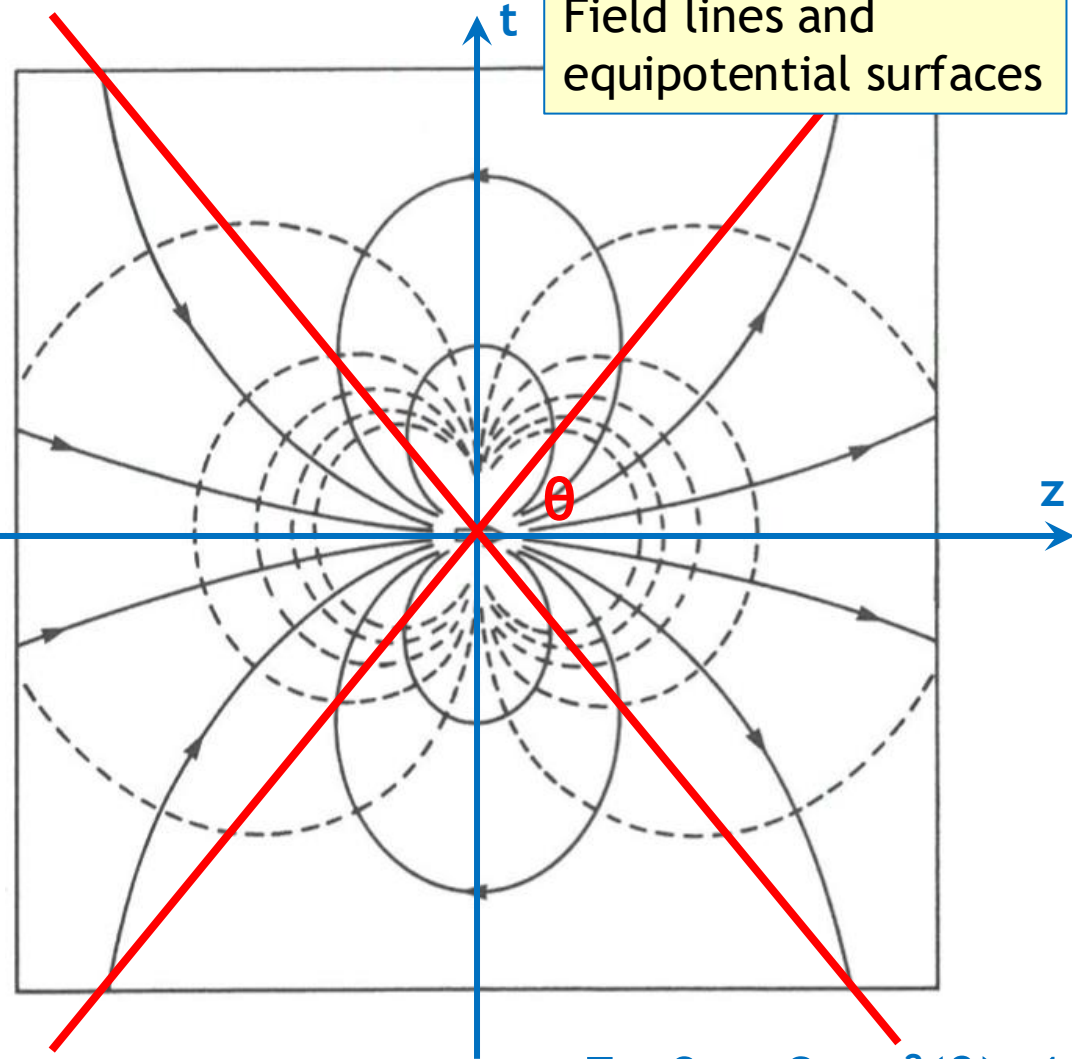


$$E_z|_{\theta=0} = -2E_z|_{\theta=\pi/2}$$

$$E_{\perp}|_{\theta=0;\pi/2;\pi} = 0$$

On the z axis: $E_z(z) = \frac{p}{2\pi\epsilon_0|z|^3}$

Field lines and equipotential surfaces



$E_z=0$ at $3\cos^2(\theta)=1$,
i.e. $\theta \approx \pm 1$ rad

Dipole in a uniform field

Force acting on a charge distribution in a uniform electric field $\vec{E}_0$:

$$\vec{F} = \int_V \rho(\vec{r}) \vec{E}_0 dV = \vec{E}_0 \int_V \rho(\vec{r}) dV = Q \vec{E}_0$$

Torque exerted on a charge distribution in a uniform field:

$$\vec{\tau} = \int_V \vec{r} \times (\rho(\vec{r}) dV \vec{E}_0) = \int_V (\rho(\vec{r}) \vec{r} dV) \times \vec{E}_0 = \vec{p} \times \vec{E}_0$$

For a dipole ($Q=0$), there is *only* a torque, as long as the dipole is not parallel to the external field.

Potential of a uniform field $\vec{E}_0$: $\varphi(\vec{r}) = -\vec{E}_0 \vec{r} + C$

Proof (*a non-assessed problem*): $-\nabla \varphi(\vec{r}) = -\nabla(-\vec{E}_0 \vec{r}) = \vec{E}_0$

Potential energy of a dipole in a uniform external field:

$$U = \int_V \varphi(\vec{r}) \rho(\vec{r}) dV = \int_V (-\vec{E}_0 \vec{r}) \rho(\vec{r}) dV = -\vec{p} \cdot \vec{E}_0$$

Minimal energy: $\vec{p}$ in the direction of $\vec{E}_0$.

The dipole moment tends to align with the external field.

Dipole in a non-uniform field

A non-uniform field, approximated by first-order terms in the Taylor expansion about the origin:

$$\vec{E}(\vec{r}) = \vec{E}_0 + x \frac{\partial \vec{E}}{\partial x} + y \frac{\partial \vec{E}}{\partial y} + z \frac{\partial \vec{E}}{\partial z} = \vec{E}_0 + (\vec{r} \nabla) \vec{E}$$

Force acting on a charge distribution placed at the origin:

$$\vec{F} = \int_V \rho(\vec{r}) \left[\vec{E}_0 + (\vec{r} \nabla) \vec{E} \right] dV = Q \vec{E}_0 + (\vec{p} \nabla) \vec{E}$$

Force acting on a dipole ($Q=0$): $\vec{F} = (\vec{p} \nabla) \vec{E}$

Easier to understand considering individual components, e.g.

$$F_x = (\vec{p} \nabla) E_x = p_x \frac{\partial E_x}{\partial x} + p_y \frac{\partial E_x}{\partial y} + p_z \frac{\partial E_x}{\partial z}$$

Two coaxial point-like dipoles at a distance x : 

$$F_x = p \frac{\partial}{\partial x} \left(\frac{p}{2\pi\epsilon_0 x^3} \right) = -\frac{3p^2}{2\pi\epsilon_0 x^4}$$

This is a specific type of van der Waals attraction.

Summary

- ❖ Electrostatic potential of a point-like electric dipole:

$$\varphi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3}$$

- ❖ Electric dipoles in uniform fields: zero net force; non-zero torque; dipole moments tend to align along the field lines.

$$\vec{\tau} = \vec{p} \times \vec{E} \quad U = -\vec{p} \cdot \vec{E}$$

- ❖ Electric dipoles in non-uniform fields: non-zero net force.
A dipole aligned with the field is attracted towards stronger field.

$$\vec{F} = (\vec{p} \cdot \nabla) \vec{E}$$

- ❖ Electrically neutral objects with non-zero dipole moments ($\mathbf{Q=0}$, $\mathbf{p \neq 0}$), e.g. water molecules, experience electrostatic forces.